

Sensor Platform

IG Handle is the physical acquisition boundary for GRANDE. It records what is installed, how each device is reached, which frame and topic it owns, how time is stamped, and whether the device is currently available.

The deployed inventory is defined by `sensor_contract.yaml`, not by launch-file names. `sensor_network.yaml` records runtime endpoints and interface roles. `sensor_frames.yaml` records configured static transforms and their evidence state, while `sensor_models.yaml` records capabilities independent of one deployed serial.

Changing a physical transform requires measurement evidence. Runtime code may select a configured device but cannot patch its geometry or identity.

The host uses separate local and boat-facing network roles. Stable device identity uses serials, USB attributes, udev aliases, and explicit network endpoints. A changing `/dev/ttyACM*` number or switch port is not itself a new identity. The runtime network contract places the sonar endpoint and host at `192.168.2.4` and `192.168.2.10`; the checked-in `01-netplan.yaml` template now uses that same `192.168.2.0/24` subnet. Applying netplan remains an explicit operator action after verifying switch wiring, endpoint identity, and link state. The Heron-facing host address is `192.168.131.10`, matching the deployed networkd configuration; the base computer remains `192.168.131.1`.

Camera serial files reserve one calibration record per physical camera. The current records are explicitly uncalibrated placeholders and must not be treated as metric camera calibration. The sensor contract assigns ordered inspection roles so DEFECTOR and ORACLE consume role, topic, frame, and calibration state without product-name coupling.

Device stamps are preserved when available. IG Handle publishes timing inputs for downstream estimation and records raw sensor evidence without turning acquisition health into mission or actuator authority. GRANDE normally owns integrated recording; IG Handle can collect isolated raw evidence for hardware investigation.

IMU Ownership and Recovery

GRANDE's canonical physical IMU is the serial-qualified Xsens MTi-30 on IG Handle. Sensor contract ID 1 binds serial `0368319D` to `/dev/sensors/imu`, publishes only below `/sensors/imu`, and is owned by the persistent `ig-handle-xsens.service`. The service waits without opening another device when the exact serial or physical Heron ROS master is unavailable. It restarts the provider after USB loss, stale output, driver exit, or ROS-master replacement and refuses duplicate publishers. `sensor_bringup` observes this external owner for readiness but never opens or stops its serial port.

The stock Heron `/imu/*` and `/cv5/ros_msc_l_node` surfaces describe an optional onboard MicroStrain installation. They are not aliases for the IG Handle Xsens. No Xsens topic is bridged into `/imu/data_raw`, and Xsens recovery grants no navigation or actuation authority. A Heron without the optional onboard sensor may retain inactive stock diagnostics, but GRANDE readiness uses the canonical `/sensors/imu/data` contract.

The Teensy timing design and its disabled-by-default electrical gates are specified in `sensor_timing.md`. Host camera, LiDAR, and IMU acquisition remains continuous; the firmware's trigger scheduler is a future commissioning surface rather than an enabled runtime claim.

Battery Identity and Telemetry

The platform inventory distinguishes one fixed compute battery from two interchangeable propulsion batteries. `IGHANDLE-01` is commissioned against the complete live JK BMS identity: BLE address, device name, model, hardware and software versions, serial number, and manufacturing date. Voltage, current, and charge behavior are measurements and never serve as identity. A host-wide lock prevents concurrent processes from owning the same BLE device, while periodic device-information queries re-establish that identity during continued telemetry acquisition. The `/ighandle_jk_bms` node publishes standard state on `/sense_ighandle`; its paired

`/sense_ighandle/details` record carries identity, raw pack and cell fields, status, alarms, counters, and provenance with the same timestamp. Fleet admission requires both synchronized messages to retain that exact caller identity; another publisher cannot establish compute-pack identity.

`HERON-01` and `HERON-02` represent physical propulsion packs that the Heron MCU cannot identify electrically. They remain uncommissioned until the matching durable labels are physically present and inspected. After commissioning, GRANDE binds a labelled installation to physical `/sense_heron` observations through an append-only event ledger. A new logger session, publisher restart, sequence or timestamp rollback, or source-progress gap prevents an older selection from carrying forward without explicit reconfirmation. The resulting effective assignment is published on `/battery/heron_identity` and recorded beside the measurement streams. This separation allows both removable packs to build independent histories while keeping unassigned observations visible as `UNKNOWN`.

The installation ledger attributes selection and removal to the local `username@hostname` caller by default. A supplied `selected_by` value is retained as operator-entered provenance, not as authenticated user identity.

Persistent selection state, installation history, normalized samples, and figures default to `~/.local/share/grande/battery`. The standalone IG Handle service uses a local ROS master when the Heron is off; the combined service uses the physical Heron master and validated `/sense_heron` ingress. Both use the same data root and are mutually exclusive. Repository data remains retained historical evidence rather than live mutable state. Energy integration is available only from contiguous, identity-qualified compute pack power. Propulsion motor-controller currents are preserved for diagnosis but do not establish total pack current; propulsion energy therefore remains null with an explicit unavailability reason.

Imaging Sonar

IG Handle separates physical sonar identity and acquisition from downstream mapping and mission use. The configured legacy Imagenex/DeltaT path and the candidate Ping360 path are different providers; their protocols and geometry are not interchangeable. The current sensor contract deliberately leaves sensor 8 provider-unverified and selects passive `udp_raw`; it does not claim that the attached head is a commissioned DeltaT or Ping360.

The DeltaT wrapper launches the vendor executable and forwards every UDP datagram in `SonarRawPacket`, preserving receipt time, source endpoint, packet kind, provider, model, extrinsic revision, and sequence; the low 32 sequence bits are also placed in the ROS header for downstream trace correlation. The physical DT100-labelled path is fail-closed: its contract keeps `hardware_commissioned: false`, so neither the vendor process nor its raw receiver is started. Promotion requires a verified numeric UDP source IP from the beamforming/output computer; an optional nonzero source port makes the admission check cover the complete endpoint. The sonar-head control address is not substituted for this output-source identity. Packets from every other source are dropped before they acquire DT100 provider provenance. Passive `udp_raw` capture remains available separately and stays labelled `unverified_udp`. MARINER accepts only the documented 83P profile-point format and rejects 83A, 83B, 837, malformed lengths, and invalid headers. Its selected profile additionally requires 480 beams over a 120-degree sector, a -60-degree first beam, 0.25-degree spacing, 5000-sample high-resolution processing, intensity output, 240 kHz, and the selected sound-speed value. These are wire-admission settings, not proof that the vendor process was configured correctly; a mismatch publishes no cloud. It does not reinterpret other formats as a shared byte layout. The checked-in Ping360 provider implements UDP transport only; USB or serial transport is not implemented and must not be assumed. When selected, the UDP provider uses the Blue Robotics Ping protocol and publishes both raw messages and a provider-neutral profile carrying angle, range resolution, gain, frequency, source identity, acquisition frame, and extrinsic revision.

Physical endpoint, identity, frame, and profile live in IG Handle. MARINER may consume an accepted profile for mapping; ORACLE may request a sonar-relevant mission; neither owns device commands.

DT100 and Ping360 have separate configured frames. The DT100 seed is a down-looking 120-degree cross-track fan on `dt100_link`; Ping360 is a horizontal mechanical scan on `ping360_link`. Their current revision tokens (`dt100-seed-2026-08-11-v1` and `ping360-seed-2026-08-11-v1`) identify separate configuration seeds, not measured calibration. Both transforms remain physically unverified, so marker observations are shadow-only until the active provider, axes, origin, and revision are measured. For a structured

marker, DT100 is the primary pose sensor when its simultaneous fan intersects the constellation. Ping360 is a useful 360-degree discovery or planar fallback, but its sequential sweep and broad vertical aperture do not supply an instantaneous six-degree-of-freedom pose.

Read-only identity is the lowest-risk first connection and still exchanges Ping protocol request and response packets. Active acoustic scanning requires both scan mode and the configured allow flag because a scan changes physical device state. The provider rejects invalid message lengths, checksums, ranges, profiles, and identity. These runtime guards remain even without standalone validation utilities.

The simulator can publish the same canonical profile with an explicit producer- set `synthetic=true` field; the physical provider always sets it false. Replay can reproduce transport and mapping behavior. Neither establishes real acoustic propagation, multipath, target reflectivity, beam geometry, or latency. Adding this producer-set field on 2026-08-11 intentionally changed the ROS 1 `SonarProfile` MD5 to `c60a9cd87d90490ea37c2ae5164e2b76`. Older profile bags are wire-incompatible with the current message and require an explicit, reviewed migration bridge or conversion rule before replay; they must not be silently treated as current profiles.

Field evidence retains device identity, profile, sound-speed assumption, pose and frame relationship, environment, raw packets or profiles, and synchronized navigation state. ROS receipt time plus 83P latency fields can recover the packet-reported center-ping time, but no real capture has yet established the beamforming computer's clock basis, latency semantics during live output, or network-delay bound. A sonar image is observation evidence, not a metric defect measurement without reviewed geometry and calibration.

Heron Propulsion

IG Handle owns the physical facts needed to inspect and commission the Heron's propulsion hardware. MARINER owns controller mathematics and the final guarded command path; GRANDE owns evidence lineage and run-level evaluation.

The hardware chain includes battery protection, power distribution, vehicle controller, motor-enable/heartbeat path, Castle ESCs, motors, impellers, ducts, wiring, and telemetry sensing. Reverse asymmetry may come from configuration, electrical delivery, motor/ESC condition, mechanics, water loading, weight distribution, or command processing; one observation does not identify cause.

Restrained-air current can isolate electrical or mechanical asymmetry from water flow, but it does not identify afloat thrust. Current is not treated as force without an independent force measurement.

Castle Link can read and compare ESC profiles, including direction, endpoints, braking, timing, and current-related settings. The two sides should be compared as exact exported profiles before changing parameters. Physical inspection covers connectors, motor/shaft freedom, impeller and duct condition, water ingress, and side-to-side wiring differences.

ROS controller gains affect software command response but do not correct an ESC, motor, impeller, or hull-balance defect. Hardware and software changes are isolated and recorded separately.

Useful evidence aligns requested command, accepted drive, motor-enable state, voltage, side-specific current, RPM or PWM when available, and vehicle motion. Forward/reverse, left/right, coast, dry, restrained, and afloat regimes remain labelled. Replacement screening is a candidate record, not compatibility approval.